Aqueous solutions of ethanol can be produced when sugar solutions are fermented using yeast. The fermentation method is used to make alcoholic drinks. Fruit juices, such as grape juice, contains a source of sugar glucose ($C_6H_{12}O_6$). When yeast is added it feeds on the sugar in the absence of oxygen to form wine (a solution of ethanol) and carbon dioxide.
First sucrose is converted to glucose and fructose by invertase. Then glucose or fructose gets converted to ethanol in presence of zymase.

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{\text{Invertase}} \underset{\text{Glucose}}{C_6H_{12}O_6} + \underset{\text{Fructose}}{C_6H_{12}O_6}$$

$$C_6H_{12}O_6 \xrightarrow{\text{Zymase}} 2C_2H_5OH + 2CO_2$$

9. In which of the following medicine production, ethyl alcohol can be used?

(a) Antiseptic (b) Antipyretic
(c) Anti-allergic (d) More than one of the above
(e) None of the above

BPSC (Tre-3) {Class 9-10}–2024

Ans. (a)

Ethyl alcohol, also known as ethanol, can be used in the production of various medicines. One of the common uses of ethyl alcohol in medicine is in the production of Anticeptics. It is commonly used in hand sanitizers, disinfectants, and other antiseptic products.

C. Polymer

1. Natural rubber is a polymer of

(a) Isoprene (b) Styrene
(c) Vinyl acetate (d) Propene
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020
65th B.P.S.C. (Pre) 2019

Ans. (a)

Polymerization is the process of joining together a large number of small molecules to make a very large molecule. The reactants (i.e. the small molecules from which the polymer is constructed) are called monomers and products of the polymerization process are called polymers. Natural rubber is the natural polymer of isoprene. Isoprene is a colourless liquid made by destructive distillation of petroleum.

2. When sulphur is heated with rubber, the process is commonly known as

(a) saponification (b) galvanization
(c) sulphonation (d) vulcanization

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (d)

When sulphur is heated with rubber, the process is commonly known as vulcanization. It is a chemical process used to harden natural rubber.

3. Teflon is a polymer of which of the following monomers?

(a) Tetrafluoroethylene (b) Vinyl chloride
(c) Chloroprene (d) Acetylene dichloride
(e) None of the above/ More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

Polytetrafluoroethylene commonly known as Teflon is a polymer of tetrafluoroethylene.

4. Bakelite is formed by the condensation of :

(a) Urea and formaldehyde
(b) Phenol and formaldehyde
(c) Phenol and acetaldehyde
(d) Melamine and formaldehyde
(e) None of the above/ More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Bakelite is a thermosetting phenol formaldehyde resin, formed from a condensation reaction of phenol with formaldehyde. It was developed by the Belgian-American chemist Leo Baekeland in 1907.

D. Organic Acids

1. Lemon is citrus due to –

(a) Hydrochloric acid (b) Acetic acid
(c) Tartaric acid (d) Citric acid

39th B.P.S.C. (Pre) 1994

Ans. (d)

Lemon contains mainly citric acid ($C_6H_8O_7$) which fulfills the deficiency in the body. Citric acid is a weak organic acid found in citrus fruits. Citric acid is most concentrated in lemons and limes, where it can comprise as much as 8% of the dry weight of the fruit. Acetic acid is found in vinegar, while tartaric acid is found in tamarind.

2. Which acid is present in tamarind?

(a) Methanoic acid (b) Tartaric acid
(c) Lactic acid (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Tartaric acid is the major acid present in tamarind pulp which gives the pulp acidic taste. Tartaric acid is a white crystalline organic acid that occurs naturally in many plants, most notably in grapes.